Math 340 • Fall 2026 • Victor Matveev • **Lab Project #9**

5.2. Numerical Integration: Simpson's rule and Richardson's Extrapolation

PART 1: Write a MATLAB function that approximates a definite integral $\int_a^b f(x)dx$ numerically, using the composite **Simpson's integration** rule with $\boldsymbol{n}$ partitions (where $\boldsymbol{n}$ is an even integer) of the integration interval, [a, b]:

$$S_n = \frac{h}{3}\left[f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+\ldots+2f\left(x_{n-2}\right)+4f\left(x_{n-1}\right)+f(x_n)\right],\ \text{where } h=\frac{b-a}{n},\ \ x_k = a+h\cdot k,\ \ k=0,1,\ldots,n.$$

This MATLAB function should accept the following **arguments**: the function handle $\boldsymbol{f}$, limits of integration $\boldsymbol{a}$ and $\boldsymbol{b}$, and the (even) number of integration interval partitions, $\boldsymbol{n}$.

The function should only **return** one number: the approximate integral value, $\boldsymbol{S_n}$.

Then, run your code with $\boldsymbol{n = 32}$ partitions of the interval $[0, 1]$ to test the integration result $\int_0^1 e^x dx = e - 1.$

Compare the actual error of this computation with the theoretical error bound, which for $\boldsymbol{f(x) = \exp x}$ is given by

$$\left|Error\left[S_n\right]\right| = \left|\frac{f^{(4)}(c)}{180}(b-a)h^4\right| = \left|\frac{\exp(c)}{180}(b-a)h^4\right| \le \left|\frac{e\,(b-a)h^4}{180}\right|$$

PART 2: Systematically increase the number of partitions $\boldsymbol{n}$ to analyze the behavior of the error of the Simpson's rule, using your code from part 1. To examine the error, we will use the following integration result:

$$I \equiv \int_0^1 \exp(-x^2)dx \approx 0.746824132812427.$$

Write a piece of code that uses your Simpson's integrator code, comparing the numerical result with the high-precision numerical value $\boldsymbol{I}$ given above, within a for-loop over increasing n values given by $n = 2^k$, where $k = 1,\ 2,\ldots,7$.

For each of these values of $\boldsymbol{n}$, your code should print out the following, as a formatted table:

- The number of partitions, $\boldsymbol{n = 2^k}$ $(k = 1 \ldots 7)$.
- The numerical integration result, $\boldsymbol{S_n}$ (make sure to show 15 digits after the decimal point).
- The actual integration error, $\boldsymbol{E_n^S = |S_n - I|}$.
- The ratio of successive errors $\boldsymbol{\left|E_{n/2}^S / E_n^S\right|}$

What value does the ratio $\boldsymbol{\left|E_{n/2}^S / E_n^S\right|}$ approach as $\boldsymbol{n}$ is increased? **Explain** this result in terms of the Simpson's rule error.

PART 3: Modify your code from part 2 by including **Richardson extrapolation** of the Simpson's rule values $\boldsymbol{S_n}$. Namely, for each entry in the table starting with $n = 4$, the code should also print the following more accurate integral approximation (equivalent to what is known as the ***Boole's rule***):

$$B_n = \frac{2^4 S_n - S_{n/2}}{2^4 - 1} = \frac{16S_n - S_{n/2}}{15}$$

Your code should print a table with the following: **(1)** Simpson's rule value S_n, **(2)** its error, $E_n^S = |S_n - I|$, **(3)** the error ratio $\left|E_{n/2}^S / E_n^S\right|$, **(4)** Boole's rule value B_n (starting with $n = 4$), **(5)** its error, $E_n^B = |B_n - I|$, **(6)** the corresponding error ratio $\left|E_{n/2}^B / E_n^B\right|$ (starting with $n = 8$).

Question 1: Explain the observed behavior of both of these error ratios with increasing $\boldsymbol{n}$.

Question 2: For $\boldsymbol{n = 32}$, approximately how many times smaller is the error of B_n compared to the error of S_n?